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Real Time VPN Traffic Attack Detection and Classification

N. Chidambaresh
Dept. of Electronics & Comm.
SRM Institute of Science & Technology
Chennai, India
cn7414@srmist.edu.in

S Charukkesh
Dept. of Electronics & Comm.
SRM Institute of Science & Technology
Chennai, India
cs1838@srmist.edu.in

Dhivya G
Dept. of Electronics & Comm.
SRM Institute of Science & Technology
Chennai, India
dg0056@srmist.edu.in

VN Arumbu
Dept. of Electronics & Comm.
SRM Institute of Science & Technology
Chennai, India
arumbuv1@srmist.edu.in

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***Abstract*—Virtual Private Networks (VPN) are used by many legitimate users all over the globe as it ensures data confidentiality over public networks. With the improvement in the current technologies the chances of misusing data over transmission is prevalent. The way in which VPN ensures confidentiality may however turn as a advantage for the data manipulators to indulge in fraudulent activities. VPN provides protection against Internet Service Providers (ISP) to avoid tracking user activities and it may provide undesirable benefit to manipulators by exploiting encrypted traffic patterns and thereby simulating cyber attacks. Therefore, to ensure the use of VPN only for legitimate users and to detect malicious activities of hackers in the presence of VPN requires a system that differentiates the suspicious and normal activity. This problem is addressed through our idea by designing a system to detect and classify the malicious activity (Port Scan or not) in the presence of VPN. The system employs a real time VPN attack simulation environment (Port Scan), Detection and Classification of Port Scan using Machine Learning Techniques. Port Scan is a network reconnaissance technique to probe target host to detect the open, closed and filtered ports. This port scan attack is detected and the system classifies if the attack is port scan or not. The real time VPN environment is setup in Linux to simulate normal, malicious traffic and Machine Learning Techniques are trained using a dataset to detect the malicious traffic and classify if its a port scan attack or not. Integrating the designed ML model on to a real time VPN environment will monitor the traffic network using pcap file and detect the presence of attack and classify the environment as port scan induced attack environment or not. This system thereby understands the VPN environment, the Port Scan attack and enhance VPN security for legitimate users.**

***Index Terms*—Virtual Private Network, Legitimate users, Cyberattacks, Port scan, Machine Learning Techniques.**

I. Introduction

The advent rise of the technologies over the days has drastically improved the internet's connectivity. The massive growth in today's prominent fields is due to faster internet accessibility. The request to open any website or app via internet is monitored by the ISP which witnesses the websites an individual has visited. However, while using Virtual Private Network (VPN) the entire search process is secure & encrypted. VPN server acts as a medium to tunnel the data that is searched and the website interacts with the VPN server instead of actual ISP. The ISP can now witness only the individual's connectivity with the VPN server and not the actual data or search made by the individual. VPN is a technology that will create a secure and effective tunnel between the user and the internet. This VPN server provides the benefit of Privacy to hide the IP address, location, also to encrypt and protect the data from attackers when using public network. However, attackers tend to misuse VPN for their own advantage, as the hackers use VPN server to hack other systems there are no possibility of using existing systems like Firewall, Intrusion Detection System, etc to block their activities.

This creates an added advantage for the intruders to execute various manipulation activities, hiding their IP address and safeguarding their origin. They may also switch VPN servers or accounts that increases the chance of becoming untraceable. This can adversely affect the VPN servers used by legitimate users. The security systems may block legal VPN users or their VPN traffic maybe treated as suspicious.

There are various attacks initiated by the Cyber criminals including DDoS Attacks, Phishing activities, Botnet command, Credential stuffing & Brute force Attack, etc. Some other threats include Vulnerability in VPN protocols, Misclassification of VPN traffic as intruders may disguise themselves as legitimate traffic. They may exfiltrate senstive corporate datas via enterprise VPN or use Malware over VPN by delivering malware payloads through the VPN tunnel to make the detection process harder. This creates a need to handle the intruders and protect legitimate users from any attacks in the presence of VPN. Employing deep learning techniques including autoencoders, CNNs, RNNs, DBNs can be effective for intrusion detection [1], further Encrypted traffic classification performed using Deep Learning outperformed Shallow Machine Learning approaches for mobile human data traffic [2],

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likewise Deep Packet approach is utilized for application identification and Encrypted traffic characterization [3]. A Network Intrusion Detection system for traffic classification is also developed through Deep learning for tackling challenges such as high computational overhead, imbalanced class distribution and the limitations of outdated datasets [4]. The existing VPN IDS focus on encrypted traffic classification, employing static dataset and building models with high computational requirements, our work focuses on simulating attacks within a VPN environment to derive conclusions on attack presence and classification employing a Linux based Framework, thereby monitoring the attacks and analyzing its pattern for further security enhancements. As the VPN protects the data, the detection of malicious activities pose difficulty, for which alternatively the packet sizes are studied, timing and patterns are understood to analyze the packet. Classification algorithms are developed with Machine learning models to detect and identify abnormal VPN traffic, unusual traffic spikes and port usage.

A linux Based malicious detection and classification system capable of identifying whether the traffic is of the attackers or legitimate users is designed, additionally the system labels whether the attack generated is Port Scan or not. The system tends to classify the detected activity as Port Scan or benign to derive effective countermeasures for the generated Port Scan attack. The system intends to understand traffic patterns in the presence of VPN and rapidly monitors the activity, thereby classifying the environment is intruded or not. This system classifies the attack features, paving the way for the development of a required solution to avoid it. This framework can be expanded for other types of attacks by simulating them within a VPN server and evaluating the system performance to improve the accuracy. This provides an effective solution to avoid the misuse of VPN servers in various enterprises, thereby upholding the privacy of legitimate users.

II. Literature survey

Virtual Private Networks (VPN) provide secure tunnel to handle private documents and data by avoiding ISP's detection. This creates an added advantage for attackers to indulge in malicious activities and it might lead to misinterpretation of legitimate users as intruders. This creates a requirement of generating solutions to detect the attack, classify it and analyze the same for initiating appropriate countermeasures. As traditional methods like Signature IDS cannot detect unknown attacks, there is a requirement of manually updating the rules. To produce high false positives, ML algorithms can be utilized in which the traffic patterns are studied and the attacks are accurately analyzed with continuous training of the model. Intrusion Detection systems are developed to effectively work against attacks and these systems employ deep learning techniques including autoencoders, CNNs, RNNs, DBNs and evaluates metrics by identifying key challenges like poor detection and outdated datasets [1]. Encrypted mobile traffic data is studied and classified using various approaches, Deep learning. acts as the classifier model which is compared and evaluated against the model built with the Shallow Machine learning, to conclude that Deep Learning outperforms the latter Machine Learning technique also highlighting the risks against biased inputs [2]. Encrypted traffic data can further be classified using Deep packet approach to achieve 98% of application identification and 94% of traffic characterization accuracy [3]. In the same way an Intrusion Detection System primarily employing deep learning is built to address the issues including class imbalance, lack of real world validation and outdated datasets [4]. Statistical techniques for encrypted traffic data classification were studied and its metrics were evaluated [5]. Network Anomalies are identified through various ML algorithms namely Neural networks, Support Vector Machines, Random Forest for improving network security [6]. Traffic identification through ML approaches improve the overall network security and efficiency [7]. Various ML models are built to analyse the traffic anomalies [8]. This Traffic detection and analysis process might improve the overall mitigation strategies to overcome other attacks categories [9]. The abnormality in traffic patterns are studied to understand the attacker behaviour in high volume networks [10]. Network traffic is handled through various techniques, its classification results are derived by employing Active learning [11] which proves to be a efficient traffic classification methodology by reducing the requirements of labeled data. In addition, Machine Learning can be applied to detect and classify various attacks like Malware, DDoS, Phishing etc. using supervised, unsupervised, ensemble and deep learning trends [12]. The Man in The Middle attack (MITM) is studied and simulated to understand its behaviour, there are various solutions to handle the problems of MITM attacks as discussed [13]. Specific attack categories and widely recognized attack types like DDoS and Botnet are further understood by employing ML algorithms and mitigated using traditional techniques, in which ML based techniques holds higher success rate compared to other methods [14] [17]. DDoS attack detection and prediction can be performed using ML algorithms by analyzing the SDN network for the feature collection process [15] and building Intrusion Detection System to detect and analyse DDoS attacks [16]. Botnet attacks are detected and mitigated using Machine Learning Algorithms for efficient network conditions [18] [19]. In most of the real time scenarios, attack detection and prediction becomes crucial and this can be achieved by integrating ML algorithms in real time conditions after training the Model with appropriate dataset to reflect the detection of attackers [20]. However, in the presence of VPN attackers tend to use the network for hacking, data stealing etc, that creates difficulty in understanding their activity under VPN tunnel. There are various cross-protocol attacks that increase the vulnerability in the network and protocol based (IPsec) under the VPN Gateway [21]. This is mitigated through specific tunneling techniques and protocol selection for VPN which are studied to understand their strengths and weaknesses [22].

ML based IDs approaches are exhaustively studied for

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encrypted, anomalous traffic, for specific attack categories and VPN specific vulnerabilities. However, very few works combine the two: validating an ML-based detector on traffic captured live, inside an active VPN tunnel, rather than on an offline dataset. The proposed system uses a controlled deployment that validates performance directly in a live VPN tunneling environment, while still employing the offline dataset for a large and diverse training base.

The primary contribution of this work includes the development of controlled, repeatable real-time VPN attack simulation environment built on WireGuard and the extraction of the flow based statistical features from the encrypted VPN traffic environment using CICFlowmeter. A two stage ML detection & classification framework is developed and validated in real-time with the live attack situations.

III. System Description

The Real Time VPN Attack Detection and Classification system is intended to normalise Virtual Private Network usage for legitimate users. Developing a system for this safe and secure form of VPN usage provides integrity and enhancement of legal user's activities. The system integrates various individual sub-systems including Attacker, Victim and Analyser, which replicates a real time model of an Intrusion Detection System that will understand the attack simulated, detect its origin and even signify whether this environment is a malicious or normal traffic. This is achieved by training the Machine Learning model with continuous and updated datasets. The system model, its working, steps of operation, execution and the Machine Learning classification are discussed briefly.

A. Overview of the system

The implemented system is optimised to detect attacks and classify the Port Scan attack environment compared to a normal environment within a VPN setup. There are various operational steps and mechanisms upon which the system works as depicted in Fig. 1. The Attack Generation layer is where the VPN tunnel is being set up and generation of Port Scan attack takes place. Data Capture and Analysis layer captures the network packets and then analyses it to extract flow based statistical features. This layer includes the victim system that experiences the attack and the analyser system that analyses the attack patterns. The ML Layer is responsible for identification and classification of the malicious VPN traffic with the Random forest- XGboost framework developed.

B. WireGuard VPN setup and Network Topology

The Fig. 1 describes that our system setup consists of three Virtual Machines connected through a private host-only network. The Attacker VM (Kali Linux) and Analyser VM (Ubuntu) are connected through a WireGuard tunnel: the Analyser exposes a wg0 Wireguard interface, the traffic generated by the attacker is encapsulated inside the WireGuard UDP tunnel before connecting with the Analyser. The Wireguard, being a lightweight and modern cryptographic design was selected to serve the purpose of tunneling. Wireguard

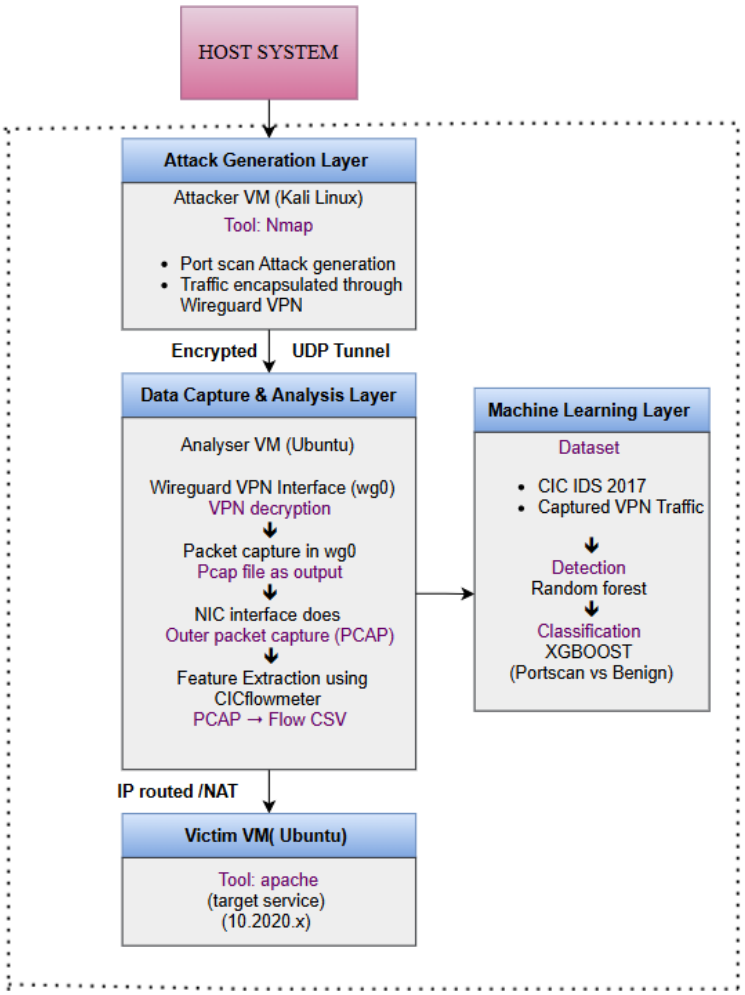


Fig. 1. Block Diagram of the System

employs dedicated /24 subnets for endpoints, a UDP listening port (51820) and default WireGuard handshake interval. The VPN decryption and packet capture is carried out on the interface producing the outer-tunnel PCAP, while the inner traffic destined for Victim VM addressed in the 10.20.20.x range is forwarded using IP routing.

The Victim VM runs an Apache HTTP server which ensures that the Analyser alone observes the encrypted traffic at the point of capture, producing the conditions of a real VPN provider. For traffic generation, the Attacker uses NMAP with multiple scan profiles including - TCP SYN scans, TCP connect scans and service detection scans with different time templates. These are exposed to the Apache service to produce Port Scan flows, however benign traffic is generated with normal HTTPS requests from the Attacker/Analyser side to both the Victim VM and to external internet hosts to maintain the dataset with a realistic assortment of internal and external benign flows along with the Port Scan Flows.

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C. Dataflow Description

The system function starts with the Wireguard VPN tunnel establishment between the Attacker and analyser system ensuring an encrypted communication between both the systems. Once both the systems are connected through VPN tunnel, a Port Scan attack is generated from the attacker system in the presence of the VPN to mimic the real-time intrusion scenario. The victim system experiences the generated attack traffic and analysed by the analyser system. The analyser system captures network packets by monitoring the traffic packets transmitted through the VPN tunnel. The captured packets are stored as PCAP files and analysed using CICFlowmeter to extract numerical and statistical features including flow duration, byte rates, flag information, packet counts and inter-arrival times. The dataset extracted then undergoes feature preprocessing stages to begin the ML classification. The developed Random forest - XGboost framework involves both the models, where the Random Forest model is employed for traffic identification, XGboost model determines whether the detected activity corresponds to a Port Scan attack or benign traffic.

D. Dataset and Features

Machine Learning Techniques are employed for this detection and classification process, in which these techniques include the responsibility of successful implementation of secure mechanism for legitimate VPN users. The ML model is trained with an existing dataset and feature extraction from the dataset helps in identifying the presence of attack and its type. CIC-IDS 2017 dataset is used for training and augmented with Newly captured VPN based traffic. Each Network flow consists of the following features such as Total Forward and Backward Packets, Flow duration, Average Packet Length, Number of unique destination ports, Packet rates (packets / sec), Flag counts, Bytes per flow etc. After the data is combined and augmented, the resulting dataset includes 956,961 benign flow records, 11,086 malicious flow records to sum up a total of 968,047 labeled flows. On preprocessing the dataset, 77 numerical flow features were extracted per record, the dataset was split into 774,453 training flows and remaining were reserved for validation and testing to preserve the (highly imbalanced) class ratio. All these 77 numerical features were retained and undefined values are assigned with the median of each feature column. The most pivotal features for differentiating the malicious and benign flows based on feature importance scores are: flow duration & inter-arrival time statistics - that capture rapid probing behaviour of the attack, packet-rate and bytes per second features - to analyse small packets in quick succession, forward/backward packet length statistics to witness large and variable payloads of a genuine flow, TCP Flag count features indicating the half-open and rejected connection attempts made by SYN-scan. These features were helpful in differentiating the attack and benign traffic and are consistent with flow-level signatures for port scan traffic.

IV. Methodology

The purpose of this system is to adopt advanced technologies to enhance the VPN tunneling process and provide legitimate access to normal users. This system's integration with various mitigation techniques avoid intruders entry in VPN presence. The system model works in a sequential way integrating the employed technologies and the complete system workflow is discussed.

A. Preparation of Environment

The work setup is depicted in Fig 2. starts with establishing a virtualised Linux environment with three virtual machines Attacker, Analyser and Victim systems. The attacker system is deployed in the Kali linux with the Nmap, a tool for network scanning. This Nmap is used for the generation of the Port scan traffic over the network. The analyser system is deployed as the central monitoring system in Ubuntu and integrates with VPN tunnel established using Wireguard. The CIC flowmeter, Python are installed in the Analyser system. The network packets passing through the VPN tunnel is captured and monitored using tcpdump and Zeek, it is then used for obtaining the packet traces for further processing. The Victim system running in Ubuntu hosts a target web service using Apache HTTP server to emulate a real time network service that could be probed during the attack simulation.

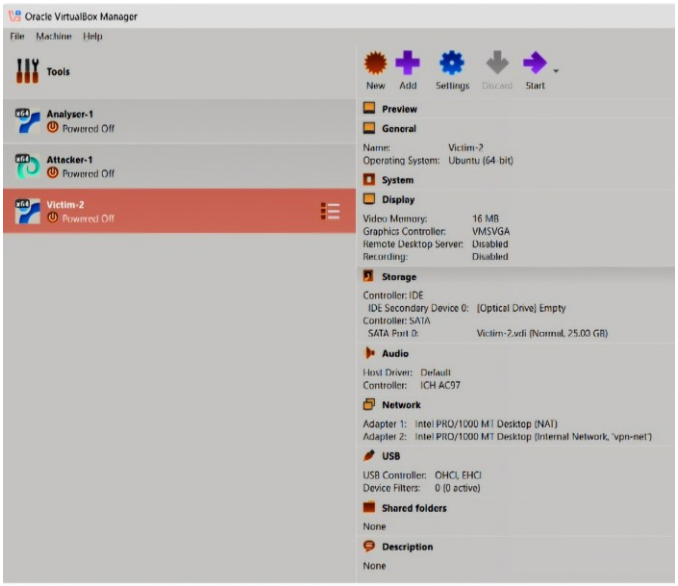


Fig. 2. Virtual Machine setup

B. Defining Attack Scenarios

At this point, controlled attack scenarios were set up to test the system in the VPN environment with malicious traffic. The main attack that was simulated in this work is Port Scan. The attacker node used Nmap to look for services on the victim machine. Port Scan attack was selected for the following reasons, one being the common reconnaissance

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technique used by intruders before launching a more vital follow-up attack. Second, the attack produces a distinctive flow-level signature like short flows, high packet rates and flag patterns that remain well suited to validate whether flow-based, payload-independent features are discriminative once the traffic is encrypted inside a VPN tunnel. Third, this attack allows controlled testbed, run-manifest methodology that are focused to understand the attack type and validated thoroughly before extending to further attacks.

In addition to this attack, the CIC-IDS 2017 training data includes FTP-Patator and SSH-Patator brute-force attack flows, the XGBOOST stage is trained as a multi-classifier model even though the paper focuses on a single attack category. The attack agnostic system produces the traffic and is tunneled, captured and converted to flow features. Additional attack types such as Distributed Denial of Service (DDoS), brute force login attempts, botnet and malware propagation can be reproduced through augmentation of Attacker's traffic generation tool employing the WireGuard tunnel. A run manifest is maintained to ensure the experiments are repeated. It keeps track of the run identifier, attack type, start and end times for each experiment.

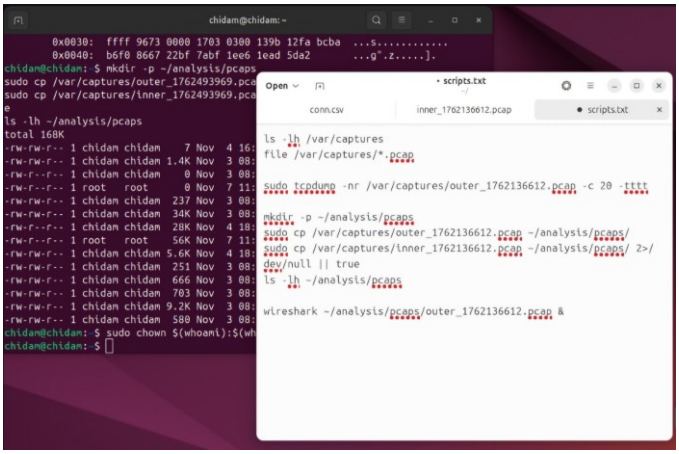


Fig. 3. Simulated Port Scan Attack

C. Generating Traffic and Extracting the flows

Based on the defined experimental runs, network traffic was generated and monitored within the VPN environment. Fig 3. depicts the simulated Port Scan attack. For each run listed in the experiment manifest, packet capture is initiated at the Analyser system. The attacker then executes the corresponding Port Scan activity according to the scheduled timestamps in the manifest. The packet capturing is prolonged until the interval ends, after which the captured traffic is stored as the Packet Capture files (PCAP). One such generated PCAP file is shown in Fig 4. These PCAP traces are further processed using CICFlowmeter to convert the packet level data into flow based records. The generated flow based data was exported as a CSV dataset containing key network attributes like source & destination addresses, packet counts, port numbers, flags, byte counts and timestamps. These features are then used for training the ML model to detect and classify the malicious traffic.

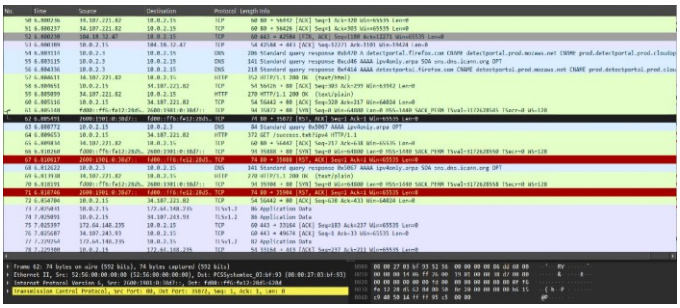


Fig. 4. Generated Pcap File

D. Feature Engineering and Dataset Preparation

The Random forest-XGboost framework was created using flow-based network traffic features extracted using the CIC-IDS 2017 intrusion detection datasets and real-time captured VPN traffic. Multiple CSV files containing benign traffic, brute-force attacks and port scanning attacks were combined to form a training dataset. During preprocessing, non-feature identifier columns such as flow IDs, IP addresses, ports and timestamps were removed to avoid data leakage and ensure that only traffic attributes of behavioral patterns were used to train the model. All other columns were encoded numerically. Incomplete and undefined values were replaced using a median-based imputation technique. The final dataset consists of numerical flow features that include packet statistics, byte counts, packet lengths, flow duration and traffic rate features. A binary label was created to separate benign and malicious traffic and attack classes were preserved to conduct a multi category classification. The Final dataset is split (70/15/15 by runs) in the following scale for training and testing using a stratified sampling technique to avoid the leakage of data.

E. Model Training and Optimization

The Machine learning Framework is built as a two-step classifying system. In the initial stage, a binary intrusion detection model was developed to differentiate between benign traffic and malicious activity. This is accomplished using a tree-based ensemble classifier called Random Forest. This works well with high-dimensional tabular network traffic data. Fig. 5 best describes the model was trained with the balanced class weights to reduce the effects of class imbalance in intrusion detection datasets. In the second stage, a multi-class classifier based on XGboost was used to classify the type of attack simulated. A label encoder was used to encode the original attack labels, which were then used to train the model to differentiate the types of attacks in the dataset. Number of estimators, tree depth and learning rate are some hyperparameters configured to enable effective learning of complex traffic patterns.

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```
Loading combined_train.csv ...
Loaded shape: (968067, 78)
Using label column: Label
Binary distribution (0=benign,1=malicious):
 y_bin
0    956961
1     11106
Name: count, dtype: int64
Number of numeric features selected: 77
Train/test sizes: (774453, 77) (193614, 77)

Training RandomForest (binary detector)...
Binary detection results:
Accuracy: 0.9998915367690353
              precision    recall  f1-score   support

      BENIGN       1.00      1.00      1.00    191393
   MALICIOUS       1.00      0.99      1.00      2221

    accuracy                           1.00    193614
   macro avg       1.00      1.00      1.00    193614
weighted avg       1.00      1.00      1.00    193614

ROC AUC: 0.9999986720267167

Training XGBoost (multiclass attack-type classifier)...
Detected multiclass labels: [np.str_('Benign'), np.str_('FTP-Patator'), np.str_('PortScan'), np.str_('SSH-Patator')]

Multiclass test accuracy: 0.9999638455896784
              precision    recall  f1-score   support

      Benign       1.00      1.00      1.00    191393
 FTP-Patator       1.00      1.00      1.00      1155
    PortScan       1.00      0.99      1.00       394
 SSH-Patator       1.00      1.00      1.00       672

    accuracy                           1.00    193614
   macro avg       1.00      1.00      1.00    193614
weighted avg       1.00      1.00      1.00    193614


Saved: rf_binary_detector.joblib, xgb_multiclass.joblib, label_encoder.joblib, feature_cols.json
```

Fig. 5. Training Summary of the Model

F. Real-Time Evaluation and Analysis

Using a streaming loop that constantly takes in new flow records, updates per-source buffers, applies the predefined preprocessing steps and runs the trained classifiers for attack detection, a real-time detection pipeline was added to the analyzer system. Based on set thresholds, the system makes predicted labels that show when malicious activity is happening and keeps logs for each prediction. The Framework is tested by sending live attacks from the attacker system at different levels of intensity and the network traffic was monitored. Performance metrics such as detection accuracy, per-attack evaluation measures, detection latency and throughput were kept in track. The results were put into organized tables, graphs and the failure rate was figured out by analyzing the wrong predictions. The extracted traffic characteristics and observed attack patterns also help in making better datasets and would make it possible to use signature-based intrusion detection and mitigation systems.

V. Results and Discussion

The trained Machine learning framework is evaluated with generated VPN flow data to assess its ability to differentiate between benign traffic and malicious Port Scan attack. The model was evaluated with augmented and live VPN streams, on which the augmented dataset with a precision and recall of 1.00 for benign class and 1.00/0.99 for malicious class and the Downstream XGBoost classifier applied on the flows already flagged as malicious, achieved an overall test accuracy of 99.996%. This correctly distinguished the Port Scan, FTP-Patator and SSH-Patator flows with precision, recall and F1-score of 1.00 across all three classes as in Fig. 5. The results show that for a large augmented dataset the flow level features are highly separable for both benign/malicious class.

The trained ML framework was evaluated on flows captured from live WireGuard testbed to assess its ability. The prediction results summarized in Fig. 6 shows that flows directed to the Internal host 10.0.2.3 were mostly classified as Port Scan, while flows directed to external servers were identified as benign. This signifies that the ML framework effectively captures attack specific traffic patterns. The model achieved an overall accuracy of 75%. For the Benign class, the obtained precision is 1.00 and the recall rate of 0.67 was recorded, the achieved F1 score of the model is 0.80. While for Port Scan attack, a precision of 0.50, recall of 1.00 and F1-score of 0.67 is achieved as depicted in Fig 7. The confusion matrix shows out of 9 benign traffic flows, 6 were correctly identified as benign and 3 were misclassified as Port Scan.

```
Loaded data with shape: (16, 18)
 Models trained successfully on synthetic baseline patterns.

=== CLASSIFICATION RESULTS ===
   Protocol      SrcIP           DstIP Binary_Prediction Attack_Type
0       TCP  10.0.2.15  172.64.148.235            BENIGN      BENIGN
1       TCP  10.0.2.15    3.233.158.26            BENIGN      BENIGN
2       TCP  10.0.2.15   34.107.221.82            BENIGN      BENIGN
3       TCP  10.0.2.15   34.107.221.82            BENIGN      BENIGN
4       TCP  10.0.2.15    104.18.32.47            BENIGN      BENIGN
5       UDP  10.0.2.15        10.0.2.3            BENIGN      BENIGN
6       UDP  10.0.2.15        10.0.2.3         MALICIOUS    PortScan
7       UDP  10.0.2.15        10.0.2.3         MALICIOUS    PortScan
8       UDP  10.0.2.15        10.0.2.3         MALICIOUS    PortScan
9       TCP  10.0.2.15   34.107.243.93            BENIGN      BENIGN
10      UDP  10.0.2.15        10.0.2.3         MALICIOUS    PortScan
11      UDP  10.0.2.15        10.0.2.3         MALICIOUS    PortScan
12      UDP  10.0.2.15        10.0.2.3            BENIGN      BENIGN
13      UDP  10.0.2.15        10.0.2.3            BENIGN      BENIGN
14      UDP  10.0.2.15        10.0.2.3         MALICIOUS    PortScan
15      UDP  10.0.2.15        10.0.2.3            BENIGN      BENIGN

Results saved to: /content/vpn_attack_predictions.csv
```

Fig. 6. Live Detection and Classification Results

```
Test data loaded: (12, 7)
🔹 Realistic Accuracy: 75.00%

=== Classification Report ===
              precision    recall  f1-score   support

      Benign       1.00      0.67      0.80         9
    PortScan       0.50      1.00      0.67         3

    accuracy                           0.75        12
   macro avg       0.75      0.83      0.73        12
weighted avg       0.88      0.75      0.77        12
```

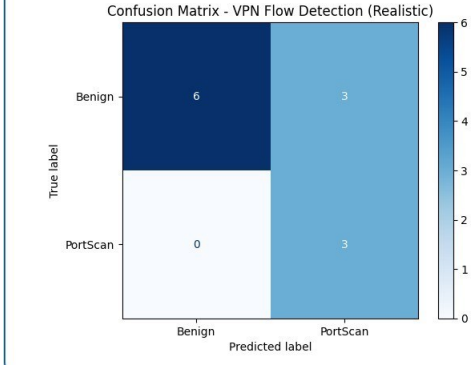


Fig. 7. Confusion Matrix

The augmented dataset based ML classification and real time classification shows significant difference. These results show a recall-based trade-off, however every Port Scan flow in live stream were correctly flagged at the cost of three benign flows being misclassified as malicious. It is evident that the model detects attack traffic with high recall though some false positives remain. This trade-off is acceptable in security applications, where detecting all the potential attacks is much prioritized and can be improved further through feature refinement and model

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tuning.

VI. Conclusion

The Real time VPN attack detection and classification system is designed and implemented as a secure encrypted network communication. This system creates a controlled VPN environment between Attacker and Analyser, the system successfully simulated the Port Scan attack. The Analyser captures and processes the VPN traffic, extracting the flow level features which in turn are analysed through Machine Learning models including Random Forest Classifier and XGBoost which detects and classifies if there is an attack or not. The system achieved reliable results by demonstrating that payload-independent, flow based features are effective for detecting Port Scan activity within VPN. While the offline Random Forest and XGBoost models achieve near-perfect accuracy on large augmented dataset, the live stream evaluation achieves 75% accuracy with perfect recall on Port Scan class. The principal limitations of the work include small live stream evaluation set limiting the live metrics statistically, only a single attack was exercised although offline models support FTP-Patator and SSH-Patator classification, the current pipeline classifies flow independently that reduces false positives. WireGuard testbed is a controlled virtual environment and has not been valid on a production scale VPN traffic volume. The future work focuses on expanding the live-capture for longer durations and multiple attack scenarios to obtain robust live-stream metrics, Extension of attack generation layer to additional attack scenarios such as DDoS, brute force logic, botnet command and malware propagation thereby developing an agnostic design for attack detection and classification, Incorporating temporal features like per-source sliding window aggregation to reduce false positives and explores deep learning sequence as an alternative to tree based ensemble. We intend to emphasize an automated Cybersecurity system that is capable of identifying attack behaviors constantly and help in deriving more solutions to tackle these intrusions in a VPN environment. This proposed approach contributes towards an intelligent, privacy preserving Cybersecurity system and can further be expanded as multi-class attack detection system with improvements in detection accuracy using Deep Learning algorithms.

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